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Writing

Putting It All Together: The Logic Behind the Forensic Scientific Method and the Inferential Test

During the last several years, I have published several articles on this website regarding the application of inference to forensic analysis¹, (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8267917105115211ae02e3>)², (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9>)³, (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20082711175115211ae2a47>)⁴ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20568953415115211ae39e6>). Although there are a few references to logic in these articles, there is no detailed treatment of the logical principles that support the methods described in them, nor is there any detailed explanation of how the methods utilized commonly by many forensic scientists and pathologists are logically fallacious. The purpose of this article is to provide an explanation of the formal logic behind what I have written previously.

I am not a professionally trained logician. Much of what I have learned has been through self-study. For this article, I utilized a logic textbook recently published on the internet (/978-0-07-731591-7?gclid=CMe_s_XB2qUCFQlubAodiBBNIw)⁵ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn12305630044d2c7ae59ae94>). For those interested in reading further, page numbers from this e-textbook appear in brackets next to several of the concepts. This allows the reader of this article easy reference to the textbook because both items can be open at the same time on the computer monitor.

Some Basic Concepts

Logic is "the study of methods for evaluating whether the **premises** of an **argument** adequately support its conclusion" [p. 1]. An argument is "a set of statements where some of the statements are intended to support another" [p. 1]. A **conclusion** is "the claim to be supported," and the **premises** are "the statements offered in support" [p. 1]. A **statement** is "a declarative sentence that is either true or false" [p. 2]. Each of the issues in a forensic analysis can be set up as a series of statements in an argument, and each argument can be evaluated for **validity** and **soundness** if it is

a **deductive** argument [pp. 3-7] or for **strength** and **cogency** if it is an **inductive** argument [pp. 50-56]. Both the Forensic Scientific Method and the Inferential Test for Expert Testimony are based on **deductive logic**, meaning that the "premises are intended to guarantee the conclusion" [p. 3]. I will discuss this first. **Inductive logic** only intends "to make the conclusion probable, without guaranteeing it" [p. 3]. I will discuss this last.

Deductive Logic, the Forensic Scientific Method and the Inferential Test

First, a reminder of the Inferential Test for Expert Testimony² ([/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9](https://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9)).

One can be reasonably certain if witness accounts of the past are consistent or not consistent with physical evidence in the present, but one cannot reliably surmise past events from physical evidence unless there is only one plausible explanation for that evidence.

Note the first part of the Inferential Test:

One can be reasonably certain if ...

If the conclusion of an analysis can be guaranteed, then one by definition is able to be certain (to know for sure, to conclude beyond doubt). Deductive inference guarantees that one can be reasonably certain (certain after the use of one's reasoning), providing that the argument is valid. A valid argument is "one in which it is necessary that, if the premises are true, then the conclusion is true" [p. 4]. One way of ensuring a valid argument is to utilize a valid argument form [p. 16]. Two famous valid argument forms are modus ponens [p. 15] and modus tollens [pp. 19, 20]. The use of these forms guarantees that the conclusion will be true if the premises are true. Modus ponens is:

If *p*, then *q*; *p*; therefore, *q*.

The italicized letters *p* and *q* are variables. Statements can replace each of these variables. For example:

Gunshot wound example #1: If a person presses the muzzle of a gun against his or her head and pulls the trigger, then he or she will get a contact gunshot wound in the head; Fred Phelps pressed the muzzle of a gun against his head and pulled the trigger (his wife saw it happen); therefore, Fred Phelps should have a contact gunshot wound in the head (*p* = pressing the muzzle of a gun against his head and pulling the trigger; *q* = contact gunshot wound in the head).

The first part of *modus ponens* has a **conditional statement**. A conditional statement — a statement utilizing "if" and "then" — has an **antecedent** (*p*) and a **consequent** (*q*) [p. 17]. *In a forensic analysis, the conditional statement is a scientific principle derived from the biological and physical sciences.* The items in the statement can be general:

If a person enters a crime scene, then he or she may leave evidence at the crime scene or some of the crime scene may leave evidence on him or her (Locard exchange principle).

Or specific:

If a person is infected by chicken pox as a child, then he or she may develop shingles as an adult.

In the second and third parts of *modus ponens* as applied to a forensic analysis, *p* is the witness (or *anamnestic*) evidence and *q* is the physical evidence related to the witness evidence. The scientific principle is applied to the particular case at hand, such as in the gunshot wound example above.

Now, consider the steps of the Forensic Scientific Method¹ ([/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8267917105115211ae02e3](https://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8267917105115211ae02e3)).

1. Acquisition of primary witness and other anamnestic evidence

2. Anticipation of future questions
3. Acquisition of physical evidence
4. Comparison of consistency of alleged events (hypothesis) with physical findings, obtaining additional data as needed
5. Assessment only to a reasonable degree of scientific certainty, recognizing the limitations of science

Step 1 involves learning p . Step three involves learning q . Step 4 involves comparing p to q . The conditional that applies to the case — symbolized as $p \rightarrow q$ — is by its nature a hypothetical statement [p. 18]. The hypothesis in a case requires knowledge of p and how it leads to q . The hypothesis does not come from assessing q without knowledge of p .

Now, consider *modus tollens*:

If p , then q ; not q ; therefore, not p . Gunshot wound example #2: If a person presses the muzzle of a gun against his or her head and pulls the trigger, then he or she will get a contact gunshot wound in the head; Fred Phelps does not have a contact gunshot wound in the head; therefore, Fred Phelps did not press the muzzle of a gun against his head and pull the trigger (his wife's statement is false) (p = pressing the muzzle of a gun against his or her head and pulling the trigger; q = contact gunshot wound in the head).

The first half of the Inferential Test —

One can be reasonably certain if witness accounts of the past are consistent or not consistent with physical evidence in the present ...

—involves applying *modus ponens* and *modus tollens*, both valid argument forms of deductive logic. With *modus ponens*, the witness account is consistent with the physical evidence as long as the physical evidence is adequately explained by the witness accounts according to a scientific principle expressed as a conditional statement. With *modus tollens*, the witness accounts are not consistent with the physical evidence when the physical evidence denies the truthfulness of the witness accounts according to a scientific principle expressed as a conditional statement. As long as the premises are correct, the truthfulness of the conclusion from these two famous argument forms is guaranteed, making it a **sound argument** (a truthful argument) [pp. 8, 9].

One point has to be made very clear. The first half of the Inferential Test can guarantee if a witness account is consistent or inconsistent with the physical evidence, but *a scientist can never guarantee that a witness account is truthful*. Consider the following **truth table for the material conditional** [p. 305]:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

We will ignore the second line of the table (scientific principles for this discussion are considered "true"). Line 1 represents *modus ponens* and line 4 *modus tollens*. Now note line 3. Lines 1 and 3 indicate that p may be true or false when q is true.

Consider this next example involving a three-year-old girl:

Example with three-year-old girl: A frantic father called 911 to report that his child became suddenly unresponsive. She had defecated in her panties, and while he had her in the bathtub to clean her up, she became suddenly unresponsive. The 911 telephone operator instructed him on

how to perform cardiopulmonary resuscitation (CPR) because he had never performed CPR before. Ambulance personnel discovered the child to be in cardiac arrest. The child was later pronounced dead. At the house, panties with feces lay in the bathtub. An autopsy disclosed abrasions in the upper abdomen, tears in the liver, hemorrhages along the colon in the upper abdomen, and small-volume blood loss into the abdomen. No abnormalities of the heart or brain were discovered. Subsequently, contrary to the father's account, the little girl's slightly older brother said that the father had struck the little girl in the chest when he had discovered that she defecated in her panties. He later described in his own way the little girl's sudden unresponsiveness, the father taking the girl to the bathtub, and the CPR performed by the father.

The physical evidence in the home of fecal stained panties and the autopsy evidence of frantic CPR with improper hand placement support both the father's and the older brother's accounts, both consistent with a sudden unresponsiveness occurring in the child from cardiac arrest. Both accounts are not the same in every respect, however. One account indicates a sudden cardiac arrest from unknown causes. This does happen. The account of the slightly older brother indicates a possible sudden heart stoppage following a blow to the chest, also known as commotio cordis. The absence of a bruise in the chest does not exclude such a blow to the chest.

In other words, two different witness accounts are consistent with the same physical evidence. Which account is true and which is false? We do not know. We do know that such an event is capable of happening according to the truth table above, so we stop short of guaranteeing the truthfulness of any witness account.

The beginning of the second half of the Inferential Test—

... but one cannot reliably surmise past events from physical evidence ...

—describes the abductive inference or “backward reasoning” referred to in several of my articles³ ([/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20082711175115211ae2a47](http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20082711175115211ae2a47))⁴ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20568953415115211ae39e6>). The reason why one cannot reliably surmise past events from physical evidence is because that commits a formal logical fallacy known as the **fallacy of affirming the consequent** [p. 36].

If p , then q ; q ; therefore, p . This is a modus ponens look-alike, but instead of affirming the antecedent (p), the fallacy occurs when the consequent (q) is affirmed instead. This is an **invalid argument form** [p. 34]. Use of this invalid argument form renders an argument unsound, even if the premises are true [p. 10]. Unsound arguments due to an invalid argument form are not reliable for determining and preserving the truth [p. 5]. As such, they should never be allowed as sworn testimony from a scientific expert unless they are accompanied by terms of uncertainty² (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9>).

There are two exceptions to affirming the consequent as an invalid argument. One exception involves the use of a **tautology** [p. 332]. A tautology is a statement that is necessarily true. It can be represented as follows:

The courts do not deal with tautologies.

The second exception is when p and q are related to each other through a **biconditional statement** [pp. 306, 307]. This is also known as the **if and only if exception** [not mentioned in the textbook].

The conditional statement

If p , then q

can also be restated as

q , if p

or

p only if q [pp. 18, 19],

so stating

p if and only if q

is another way of stating

If p , then q and if q , then p [p. 287].

So (using but not listing **simplification** [p. 346] and other intermediate steps)

If p , then q and if q , then p ; q ; therefore, p .

A biconditional statement is symbolized as $p \leftrightarrow q$.

Now for the final portion of the second half of the inferential test:

... unless there is only one plausible explanation for that evidence.

This is another way of stating p if and only if q , with q being the result of the only plausible explanation, p (and there being no other explanation).

Using the example for this “unless” exception from my article about the inferential test² ([/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9](https://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9)).

Example from “An Inferential Test...”: A body would be “discovered in a wooded area with numerous stab wounds to vital organs and multiple devastating blunt force head injuries” if somebody killed that person (manner of death homicide), and somebody killed that person only if the body of that person is “discovered in a wooded area with numerous stab wounds to vital organs and multiple devastating blunt force head injuries” (manner of death homicide and no other manner of death); Mary Jones was found in such a condition; therefore, someone killed Mary Jones (p = somebody killed that person; q = a body “discovered in a wooded area with numerous stab wounds to vital organs and multiple devastating blunt force head injuries”).

Unlike the conditional statement, the biconditional statement does not represent a concept from the biological or physical sciences to be applied in a forensic analysis. It represents instead a “smoking gun” form of evidence that does not require an explanation from a scientist. This is why I stated in my inferential test article that this form of evidence is best considered by the jury as an ultimate issue once all the evidence is presented. Jurors — not scientists — should be the ones to apply this exception² ([/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9](https://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9)).

In my first article¹ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8267917105115211ae02e3>), I stated that the standard scientific method uses **falsification** and the forensic scientific method uses **verification** for **witness evidence** and **falsification** for **circumstantial evidence** (circumstantial evidence in this article means indirect evidence without witnesses but inferred from corroborating physical evidence). In order to infer deductively that the defendant is guilty in a circumstantial evidence case, the prosecutor has to rely upon the biconditional statement, $p \leftrightarrow q$, the exception indicated in the latter portion of the Inferential Test.

Consider the following **truth table for the material biconditional** [p. 306].

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The biconditional statement, $p \leftrightarrow q$, can be **falsified** (demonstrated to be false) if the truth values ("T" or "F") of p and q differ from each other. Frequently, a prosecutor may argue a case based on circumstantial evidence by stating that there is no other plausible explanation other than his or hers for the physical evidence to be what it is. The falsification of that argument involves providing another plausible explanation. The theory, $p \leftrightarrow q$, as proposed by the prosecutor (with p being the prosecutor's explanation of what took place), may be mistaken or false (there is another overlooked explanation of the event) or q may be mistaken or false (an important item of physical evidence was overlooked or misinterpreted).

Falsifying a conditional statement ($p \rightarrow q$) is another matter. Because a conditional statement in a forensic analysis often represents a scientific concept, only scientists performing experiments that control the antecedent and the consequent are capable of falsifying such a conditional statement — not the courts. Although the courts are not equipped to falsify conditional scientific statements (even though they are equipped to falsify biconditional statements), an attorney could at least demonstrate that the testifying expert's understanding and explanation of the scientific concept is mistaken.

It is important to point out, however, that most published studies concerning past event topics such as "child abuse" do not rise to the standard of being scientific concepts. This is because such studies involve past events that are not completely controlled or even known. The vast majority of scientists have never seen "child abuse" actually occurring (i.e. have never witnessed p). As such, compilations of such "experience" are merely affirming the consequent over and over again. Rather than scientific concepts, position papers of scientific organizations regarding topics such as "child abuse" are simply expressing **conventions** (conventions are ideas agreed upon by most scientists within an organization that may or may not be based on the scientific method⁶ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn16453357715115211c42792>)).

Regarding **witness evidence**, the strategy is **verification**. Biological and physical scientists use falsification in the standard scientific method to eliminate hypotheses (demonstrate that $p \rightarrow q$ is "F" rather than "T") in order to prevent committing the **fallacy of incomplete evidence** [p. 506] (an inductive fallacy — see below). The strategy of biological and physical scientists is to maintain an open mind and not to assume that the hypothesis is correct and that everything about it is known. To avoid the same fallacy, forensic scientists must rely upon verification (demonstrate that p is "T" rather than "F") so that they will avoid committing an injustice. Even though the application of *modus ponens* cannot guarantee the truthfulness of a witness account even if it is consistent with the physical evidence, the truthfulness of the account is still assumed because of the given dictum that one is considered innocent until proven guilty. Also, an eyewitness is also considered more of an authority for

what happened than a scientist because the witness actually saw what happened and the scientist did not (an inductive **argument from authority** [p. 511] — this inductive argument and others will be described below). If a witness account is initially falsified through the use of *modus tollens*, the scientist and the police investigator return to the witness to clarify the account rather than to assume that the account was complete and the circumstances surrounding the account thoroughly understood.

Arguments from Inductive Logic

There is some disagreement among logicians and other authorities as to the definitions of induction and deduction⁵ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn21408512025115211cdbc3>)⁶ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn16453357715115211c42792>)⁷ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn18723610315115211cdcc40>). One definition of induction is that it involves “the inference of a general law from particular instances”⁶ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn16453357715115211c42792>), while deduction is “the inference of particular instances by reference to a general law or principle”⁶ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn16453357715115211c42792>). The textbook⁵ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn21408512025115211cdbc3>) distinguishes a deductive from an inductive argument as between what is guaranteed as truthful (deductive) and what is probable without a guarantee of truthfulness (inductive), without the need to indicate what is general and what is particular. The authors of the textbook further argue that making such a distinction between general and particular statements for induction and deduction is a mistake, and they demonstrate this with multiple examples [pp. 507, 508]. They also make no reference to abductive inference⁸ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn2495101865115211cddbc9>)⁹ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8293716795115211cdeb69>). For the sake of the discussion to follow, I will accept the textbook definition of what is inductive and what is deductive. In the end, it really does not make any difference which way we define these terms; both definitions seem to work out with this analysis.

As mentioned previously, an inductive argument does not guarantee the conclusion but only intends to make the conclusion probable. An inductive argument cannot be valid, because only deductive arguments are valid. With an inductive argument, one can affirm the consequent — an invalid argument form — and still have a **strong argument** and not a **weak argument** [pp. 50, 51]. A strong argument — an argument that “is probable that, if the premises are true, then the conclusion is true” — becomes a **coherent argument** if the premises are true [p. 53].

The forensic scientist or pathologist has the legitimate ability — or perhaps, the permission — to reason backwards in certain circumstances as long as the scientist uses *terms of uncertainty*: probable, possible, might happen, could happen, frequently happen, infrequently happen, suspicious for, cannot rule out, often associated with² (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9>). Also, early in a case when the information comprising *p* and *q* is not known in detail, reasoning backwards as a *heuristic* (essentially “a rule of thumb”) can be useful in developing leads because what is probable in a case may lead to more information and may help to rule in and rule out contingencies³ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn20082711175115211ae2a47>).

Still, the use of inductive argument in the form of abductive inference (backward reasoning) in the courtroom setting — even when expressed with terms of uncertainty — is dubious and frankly should not be allowed, particularly when witness accounts are available for comparison with physical evidence² (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9>)⁴ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn8479755315115211ae1aa9>).

[forensic-scientific-method-and-the-inferential-test/#fn20568953415115211ae39e6](https://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test/#fn20568953415115211ae39e6)); nevertheless, the courts currently allow the free use of abductive inference. Consequently, it would be useful to analyze such an inductive form of inference in the courtroom by using inductive logic.

Before we do this, we need to understand what is meant by **probability** when it comes to forensic analysis. Inductive inference is concerned after all with what is probable and not probable, so such a discussion is important.

The textbook discusses three theories of probability: the **classical theory**, the **relative frequency theory**, and the **subjectivist theory** [pp. 545 – 551]. The classical and the relative frequency theories cannot apply to any past-event analysis because the number of possible outcomes or the number of observed outcomes — the denominators in both equations — is unknown and not subject to mathematical determination or manipulation. The subjectivist theory — where “probability is nothing more than degrees of belief” [p. 549] — is the form of probability that applies here. In this situation, the expert is essentially stating what kind of odds he is willing to give that a particular statement is true or false. For example, when an expert states that he or she is “95% sure that the injuries are from child abuse” (a statement actually made by one expert in a court case), the expert is willing to give 19-to-1 odds for child abuse if his or her bet were placed in a casino or at a horse race. When a professional expresses the willingness to make such a bet at such stakes, he expresses great confidence in his opinion. Is such confidence warranted? Would a logical analysis support this kind of confidence?

I now offer two strong arguments. The conclusion of the first strong argument is:

If eyewitnesses offer accounts that are consistent with physical evidence, then those accounts are probably truthful.

This argument is so strong that if one of the witnesses claimed that he or she did not commit a crime, he or she should never have been arrested in the first place.

Earlier, I demonstrated that with deductive logic it is not possible to state with certainty if witness accounts are truthful or not truthful but only if they are consistent or not consistent with the physical evidence. Inferring truthfulness to a witness account would essentially affirm the consequent; nevertheless, even though we cannot guarantee the truthfulness of the witness account, we can offer a strong argument for its probability.

We can use an **argument from authority** [pp. 511 – 513]. An eyewitness may not possess an advanced degree or training in a biological or physical science, but he or she possesses a special kind of knowledge due to circumstances. He or she is one of the few individuals who was present to observe the incident in question. Furthermore, the lack of knowledge in the sciences ironically but definitely substantiates his or her reliability as a **cognitive authority** (“a person or group possessing a special fund of knowledge”) [p. 511] for the particular circumstance in question. If a witness or multiple witnesses offer statements that are consistent with physical evidence, then that person or persons cannot rely on knowledge of science to form educated guesses. The consistency of the statements would instead probably come from their truthfulness.

We can use **induction by enumeration** [pp. 513, 514]. Eyewitnesses do not offer only one observation but many observations. Past events are complex, so the multiple observations are complex in their number and their order. Another way of stating this is that p does not consist of one item but multiple items in succession (p_1, p_2, p_3 , etc.). A greater number of consistent observations allows a higher probability for truthfulness because it only takes one observation to be false to make p inconsistent with q [pp. 335, 336]. Furthermore, if multiple witnesses independent of each other offer multiple observations and all of these observations are consistent with the physical evidence, then this increases the strength of an already strong argument by enumeration. Another way

to put this is to say that a large number of **confirming instances** (a confirming instance is an "instance in which an implication of a hypothesis is observed to be true") [p. 526] offered by witnesses increases the likelihood that their statements are truthful.

Finally, we can argue on the basis of **explanatory power**. The authors of the textbook in their description of scientific reasoning state, "a hypothesis has explanatory power to the extent that known facts can be inferred from it" [p. 526]. If explanations inferred from statements provided by witnesses explain phenomena observed by scientists during an autopsy or other scientific procedure, this increases the likelihood of the truthfulness of the statements. On the other hand, it has been my experience that "theories" (essentially, hypotheses) offered by prosecutors and others who characteristically believe that all defendants lie possess poor explanatory power, not only in the scientific sense but also in the common sense. In other words, the "theory" is frequently far-fetched.

Now for the conclusion of the second strong argument:

If an expert offers abductive inferences as opinions "made to a reasonable degree of medical or scientific certainty or probability" on the witness stand, then such opinions are probably incorrect (not truthful).

This argument is so strong that an expert who infers this way would serve the cause of justice better if he or she never testified.

A treatise recently published by the National Academies of Science regarding forensic science discusses the "self-correcting" nature of proper science¹⁰ (<http://www.heartlandforensic.com/writing/putting-it-all-together-the-logic-behind-the-forensic-scientific-method-and-the-inferential-test#fn4970488005115211dc83fc>). **Self-correction** involves questioning results and correcting errors and doing so as a formal and regular practice. Such self-correction currently does not exist among scientists for issues brought before a court. Instead, many experts make positive assertions on the witness stand and appeals to their own authority to do so. Having done this, they possess neither the interest nor the ability to determine if their own assertions are truthful or not. This is because such a witness who abductively infers with certainty has neither the knowledge of the limitations for what he or she is doing nor the capacity to consider carefully the accounts of witnesses who were present to see what happened (if they had such capacity, they would not have offered abductive inferences in the first place). In spite of their appeals to their own authority (or the prosecutor's appeals to their authority), they commit an **ad verecundiam fallacy** (appeal to unreliable authority) [pp. 182 – 183, 512]. No matter the experience of the expert, that experience is unreliable if it is not consistently, formally and rigorously tested by first-hand witness accounts.

On the other hand, an expert who acknowledges the limitations of his or her science, who knows how to compare witness statements to physical evidence in deductive fashion, and who knows better than to infer abductively on the witness stand has a great capacity to self-correct. Such individuals actually learn from their experience, so their experience is probably reliable for courtroom purposes.

Also, abductively-inferring experts commit a **fallacy of incomplete evidence** [pp. 506, 512]. This fallacy occurs both at the outset of a case and when a case goes to trial. Experts who abductively infer from the witness stand familiarize themselves with q but characteristically know little about p at the outset of a case, either unwittingly or by choice. This leads them to affirm the consequent consistently at the outset. Once further information and arguments are advanced regarding p (if such information or arguments are even advanced), there is little interest in changing initial impressions — perhaps for reasons of pride, arrogance or self-preservation. This leads to an unwillingness to acknowledge the information or even to evaluate it carefully with an open mind.

Finally, such abductive inferences frequently have **poor explanatory power**. There is a human tendency to oversimplify the complexity of past events. Oversimplified explanations reflect a limited capacity to explain all the events that came before in a way that makes sense. Consequently, there is

reference to general, vague notions — including diagnoses of “child abuse” and how injuries occur only with “three-story falls” or “thirty-mile-per-hour car crashes” — without any detailed or consistent explanation of how witness accounts and other physical evidence match up with their opinions.

Conclusion

I recognize that many consider the kind of analysis offered here to be too technical — something apart from their usual area of interest or expertise. I offer it nonetheless as a basis to exclude certain forms of expert evidence from courtroom proceedings, perhaps in the form of Daubert or Frye hearings. It is my hope that attorneys will take advantage of this information to attack illogic from experts in the courtroom.

It is also my hope that scientists, attorneys and judges will take the time to familiarize themselves with this material. Too many false allegations have been offered in the name of science and too many people have suffered serious damage — even life-long damage — as a result. It is time to put all of this to an end.

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